

A New Representation based on Virtual Capacitor for Virtual Synchronous Generators

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Abstract—Grid-forming inverters will play a key role in the stability and operation of the future grid. Among different grid-forming methods developed in the literature, the virtual synchronous generator method (VSG) has become very popular due to the outstanding features it adopts from synchronous generators, like supporting the grid by inertia and autonomous grid-alignment. In this paper, a different representation for VSG is proposed by introduction of a “virtual capacitor”. It is shown that while the virtual capacitor shows the same performance as VSG, it gives a different physical representation that can be leveraged for design and study of multi-inverter systems. The simulation results corroborate the validity of the proposed method.

Index Terms—Grid-forming, virtual capacitor, virtual synchronous generator, multi-inverter systems, virtual inertia.

I. INTRODUCTION

New aggressive policies have mandated integration of renewable energy sources at a much faster pace. California, for example, has set the target of 60% renewable energy penetration by 2030, and has a plan for zero-emission energy sources by 2045 [1]. Germany plans to double the solar capacity by 2030 by adding 50GW solar panels [2]. In this new paradigm, grid-forming inverters need to take the role of synchronous generators by forming the grid and keeping the voltage and frequency in range. The challenge is to develop a grid-forming strategy that can form the grid, and yet can damp out unnecessary oscillations in case of transients.

Droop control was among the first strategies that enabled grid-forming functionality while sharing active and reactive power with no communication [3]. Its main issue, however, was the lack of inertia which could lead to poor transient response and very high rate of change of frequency (ROCOF) in many scenarios [4]. Other issues such as unsatisfactory management of transients in distribution grids with low X/R have also been reported in the literature [5], [6].

Virtual Synchronous Generator (VSG) tackles part of these problems by emulation of synchronous machine (SM) equations which results in virtual inertia support [7], [8]. The swing equation emulated in VSG based methods is basically a droop control passed through a low-pass filter, therefore the ROCOF can be effectively controlled [9]. There have been different variants of VSGs proposed in the literature. Some methods are based on full implementation of SMs up to the seventh order [7], [10], [11], while some others employ only the swing

equation for synchronization [12], [13]. In [12], for instance, virtual admittance concept is used to control voltage and current. There also have been several studies where the virtual moment of inertia is dynamically changed during transients, leading to more stable and damped response [14], [15].

Recently, several studies have been made to match the behavior of SMs with inverters instead of virtually emulating them. While in SMs the grid support is provided through rotor mass kinetic energy, in inverters DC capacitor plays the same role. It has been shown in [16], [17] that by associating DC capacitor voltage to the frequency, the same characteristics as SGs could be achieved for inverters. The main issue, however, is that some hardware requirements are needed to achieve this goal, like having a boost converter that emulates current source behavior.

This new model based on inverter dynamics inspired the authors to present a new representation for virtual synchronous generators based on “virtual capacitor”. It inherits the same excellent features of VSGs, but is more aligned with inverter physics. It enables study of transient behavior of swing equation with an RLC circuit, thus simplifying the analysis and giving new insights for transient study of multi-inverter systems based on VSG. Since this new representation does not directly use notations of synchronous machines, it will be called “grid-aligning” control in this paper.

In the next section, basics of the concept will be discussed and the new model will be presented. Simulations results will be provided in Section III to verify the analysis. The conclusions will be presented in Section IV.

II. GRID-ALIGNING INVERTERS

A. Self-Aligning feature of Synchronous Generators: an explanation

A conventional synchronous generator in the power grid functions as a voltage source that monitors only its own local voltage and current, and responds to changes it senses locally. One can show that there are three distinct phases of system behavior following any major transient – sub-transient; transient; and transactive. The “sub-transient” mode is the signaling phase where individual generators are subject to a disturbance based on natural system impedances. In the “transient” mode, the controllers respond to the locally measured disturbances and act to damp oscillations and keep the system stable. In

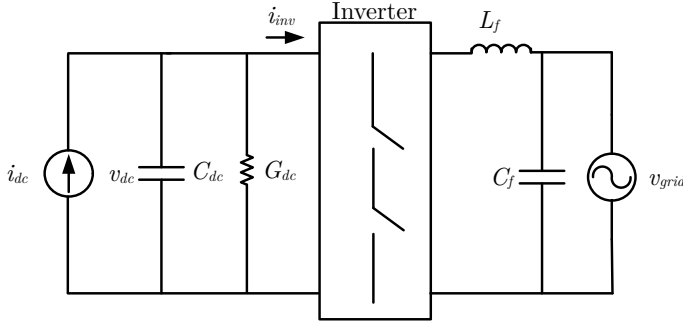


Fig. 1. Inverter structure for matching characteristics with SM

the “transactive” stage, the generators negotiate with each other and slowly move the system towards a final desired point. The physics of synchronous generators ensures that they intrinsically exchange power in a manner such that the rate of change of angle remains locked, without action from the plant controller during all different modes. This synchronizing force is referred to mostly in the literature as “synchronizing torque” [18]. Virtual synchronous generators emulate this behavior by adopting the “swing equation” which uses voltage information indirectly (through power measurement) to remain locked to the grid.

B. Matching synchronous generator characteristics with inverters

The DC capacitor acts as energy buffer in inverters and any power imbalance immediately changes its voltage. By associating the rate of change of angle, i.e. the dynamic frequency, with the capacitor voltage, it is possible for the inverter to intrinsically exchange energy with the grid to achieve a lock of the angle. This in turn, gives an inverter-dominant system ‘self-synchronizing’ capability and causes synchronizing power to automatically flow from other inverters in case of angle mismatch, forcing the angles to dynamically track each other. In other words, it allows the inverters to show the same behavioral pattern during transients as synchronous machines. The concept has been explored mathematically in [16] and here will be discussed briefly.

The power swing dynamics in an SM could be expressed by (1)-(2):

$$\dot{\theta} = \omega \quad (1)$$

$$J\dot{\omega} = -D\omega + \tau_m + L_m i_f \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}^T i_{\alpha\beta} \quad (2)$$

Where θ is the rotor angle, ω is the frequency, J is the moment of inertia, τ_m is the input torque, L_m is the mutual inductance, i_f is the excitation current and $i_{\alpha\beta}$ is the stator current in $\alpha\beta$ frame.

Similarly, for an inverter model with the structure as shown in Fig. 1, one can write:

$$C_{dc}\dot{v}_{dc} = -G_{dc}v_{dc} + i_{dc} - \frac{1}{2}m_{\alpha\beta}^T i_{\alpha\beta} \quad (3)$$

Where C_{dc} is the DC link capacitor, v_{dc} is the DC link voltage, G_{dc} is the parallel conductance and $m_{\alpha\beta}$ is the modulation signal. By setting τ_m equal to i_{dc} and v_{dc} as function of ω , the same dynamics as SM could be achieved for such as system.

The inertia constant is defined as:

$$H = \frac{1}{2} \frac{CV^2}{S_{rated}} \quad (4)$$

Which is the normalized stored electrical energy in the capacitor. G_{dc} shows how much damping the system provides in case an oscillation occurs in the system.

C. A new representation based on “Virtual Capacitor”

Matching the inverter’s characteristics is useful in a sense that it is very well aligned with inverter physical structure and makes it easier to see how much “real inertia” an inverter can provide. It inspired the authors to change the representation of VSG and to focus on “virtual capacitor” instead of “virtual moment of inertia”. As will be seen in the next subsection, it opens a path to have a different representation for the power swing transients. The value of C and R can be changed dynamically as it could result in significant improvement in transient response and frequency nadir [14], [15].

Fig. 2 shows the control structure of the proposed method. $I_{ref,v}$ could be calculated from the reference power:

$$I_{ref,v} = \frac{P_{ref}}{V_{dc,actual}} \quad (5)$$

P_{ref} and $|E|$ could be dispatch commands or the output of $P - f$ and $Q - V$ droop curves in grid-islanded mode. The virtual admittance control has been employed to emulate virtual output impedance [12].

The inverter input current (i_{inv}) can be calculated from the switching commands and inductor current. For a three-phase two-level inverter, the formula is as given in (6). It causes one sampling time delay in calculation of i_{inv} but that does not affect the performance of converter since the bandwidth of the control loops is much lower compared to the switching fast transients.

$$i_{inv} = \sum_{i=1}^3 (2S(2i-1) - 1) i_{ph(i)} \quad (6)$$

Where $S(i)$ is the switching command of i^{th} switch and $ph = \{a, b, c\}$.

D. A new framework for modeling power swing dynamics

A stable operation of an inverter requires frequency to be constant in steady state. In grid-connected mode, for instance, it means that after any transients, the inverter rate of change of angle must become the same as that of the grid. From the virtual capacitor standpoint, it is as if the grid frequency is a constant DC source, to which the virtual capacitor voltage will converge asymptotically. The following analysis will show that

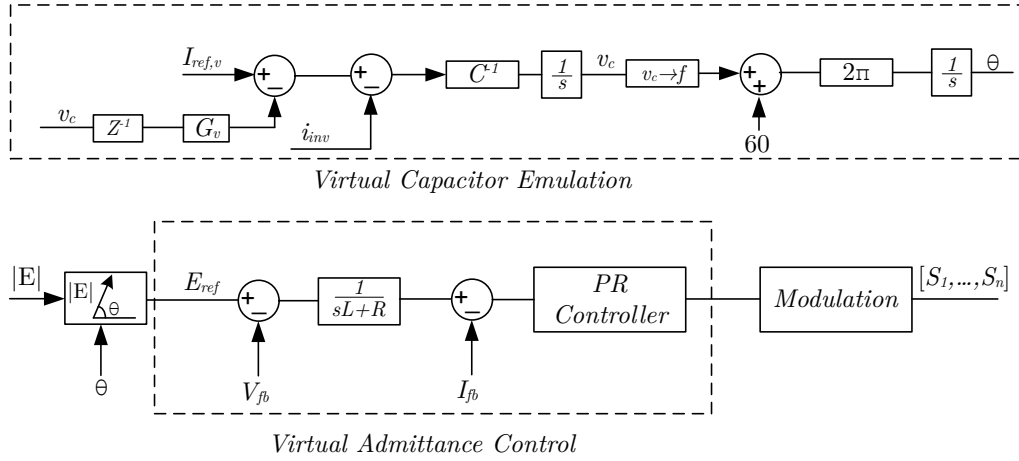


Fig. 2. Proposed Control Schematic

the dynamics during transients can be captured by adding an inductive element between the virtual capacitor and other AC sources in the system.

We assume a linear mapping between frequency and voltage, that is:

$$f_1 = k_1 + k_2 V_{dc1} \quad (7)$$

$$f_{grid} = k_1 + k_2 V_{dc,grid} \quad (8)$$

The inverter output could be written as:

$$P = \frac{3E_{inv}E_{grid}}{X} \sin \delta \quad (9)$$

Where E_{inv} is the inverter internal voltage, E_{grid} is the grid voltage, X is the inductor reactance and δ is the angle difference between the inverter and the grid.

A few assumptions are made to simplify the analysis of the power swing equation:

- δ variation is small, hence $\sin(\delta) \approx \delta$
- Inverter exchanges no reactive power with other AC sources, i.e., $E_{inv} = E_{grid}$
- Filter reactance variation due to frequency disturbances is small, i.e., $X \approx X_{nom}$
- DC source is battery, so V_{dc} is considered constant during transient.
- Capacitor filter dynamics does not affect power swing significantly.
- The inverter loss is negligible.

With these assumptions, one can write:

$$P = \frac{3E^2}{X_{nom}} \delta \quad (10)$$

I_{dc} can be calculated as:

$$I_{dc} = \frac{3E^2}{X_{nom} V_{dc}} \delta \quad (11)$$

$$I_{dc} = \frac{3E^2}{f_{nom} L V_{dc} s} (f_1 - f_{grid}) \quad (12)$$

Where f_1 and f_{grid} are inverter and grid frequency, respectively. Substituting f_1 and f_{grid} with their respective DC voltage sources in the new framework leads to:

$$I_{dc} = \frac{3E^2 k_2}{f_{nom} L V_{dc} s} (V_{dc1} - V_{dc,grid}) \quad (13)$$

Defining an inductive element L_{eff} as (14) leads us to (15) which is the power swing equation in this new framework.

$$L_{eff} = \frac{f_{nom} L V_{dc}}{3E^2 k_2} \quad (14)$$

$$V_{dc1}(s) - V_{dc,grid}(s) = s L_{eff} I_{dc} \quad (15)$$

In this new framework, the transients could be seen as interactions between virtual capacitor C_v and inductive element L_{eff} . As shown in Fig. 3, the framework allows adding higher number of inverters for transient studies. Instead of adopting full circuit model for slow transient analysis, this simplified model can be used to study interactions between different VSGs with different virtual inertia values. It can also be very useful during system level design for tuning frequency loop parameters, including virtual capacitance (C_v) and virtual resistor (R_v).

TABLE I
INVERTERS PARAMETERS (SCENARIO 1)

Parameter	Value	Parameter	Value
Nominal Power		Virtual Capacitor	
S_{n1}	10kVA	C_{v1}	6mF
S_{n2}	6.67kVA	C_{v2}	5mF
S_{n3}	5kVA	C_{v3}	4mF
Output filter values		Virtual Admittance	
L_f	6mH	L_{va}	15mH
C_f	6μF	R_{va}	2Ω
DC Voltage		Virtual Damping (Parallel resistor)	
V_{dc}	780V	R_{v1}, R_{v2}	4Ω

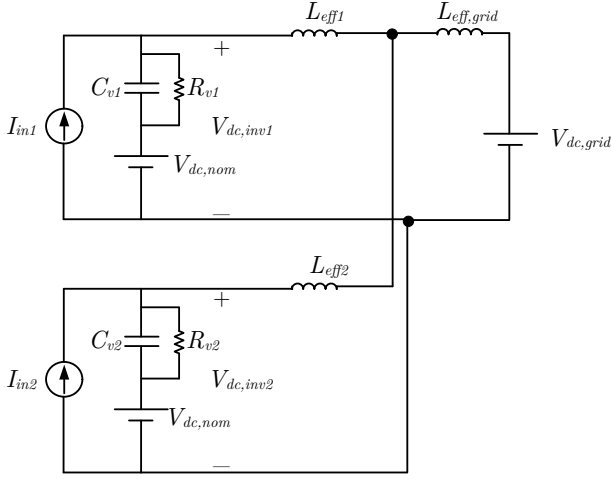


Fig. 3. Two-inverter system: Simplified model in the new framework

III. SIMULATION RESULTS

A. Evaluation of the method in the Grid-Isolated Mode

In this subsection, a scenario is defined to verify the performance of the proposed method and to elaborate more on the different modes discussed in section II-A.

A grid-isolated system consisting of three inverters and three loads have been made in MATLAB Simulink with the parameters mentioned in Table I and II. The system schematic has been depicted in Fig. 4. The reference signals for P and Q come from the external droop curves. The scenario is as follows: At $t=4s$ and $t=7s$, Load 2 and Load 3 are added to the system, respectively. Then at $t=10s$, Load 2 is dropped. The load values are reported in Table II. Fig. 5 shows the current waveform for the inverters. The sub-transient and transient modes could be clearly seen in the last sub-figure. During sub-transient, the control system does not take any action (due to bandwidth limitation) and only “observes” the transients, which in this case, are current jump and LC resonance. Then, during the “transient” mode, the control system starts its corrective action, adding damping to the system to eliminate the oscillations. Fig. 6 shows active and reactive power of the inverters. The “transactive” mode in this case, is when the sources start negotiating to share power based on a criteria which is the power droop curves in this case. In Fig. 7, it is seen that the frequency for all inverters remains locked and follow the same pattern even during sub-transients. However, it is worth noting that the self-synchronizing force highly depends on the inertia and impedance values in the system. The simulation results in total, demonstrate that the system remains stable and steady with minimum interaction during all transients.

B. Evaluation of the accuracy of the proposed model

In this part, the simplified model discussed in section II-D is evaluated. The system under test consists of two grid-connected inverters, with the same model as Fig. 3. The system parameters are the same as the previous experiment,

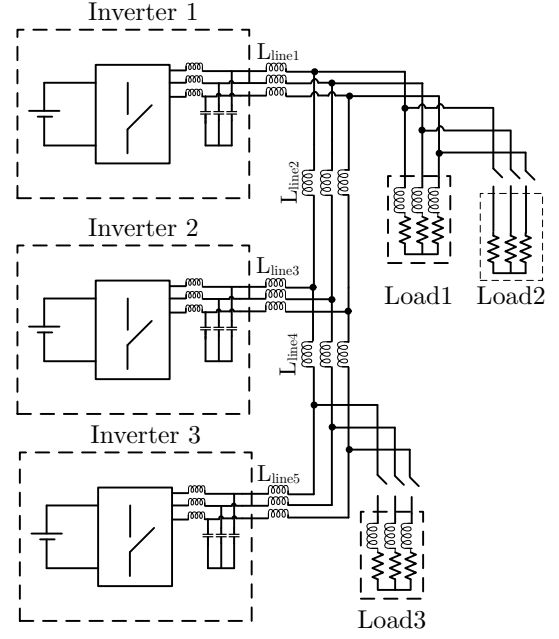


Fig. 4. Schematic of the grid-isolated system for the first scenario

except for those reported in Table III. L_{effi} and $L_{eff,grid}$ represent $(L_{fi} + L_{linei})$ and L_{grid} in the new framework. The inertia constant of the first inverter is set 6 times smaller than the second one to observe significant oscillations during the transients.

The scenario is as follows: The reference power jumps from 7.8kW to 9.4kW for inverter 1 and 2 at $t=4s$ and $t=6s$, respectively. Then, at $t=8s$, a step-wise change occur in the grid frequency from 60Hz to 62Hz. Fig. 8 and Fig. 9 show the results for inverter 1 and 2, respectively. It could be observed that the simplified model follows the real model with very low error. Also, the interactions between inverters can be captured by this model, as seen in Fig. 9 at $t=4s$. The simulation time for the full detailed model and simplified model was 96.3s and 11.2s ($T_s = 2\mu s$), showing a huge reduction in the computation time.

IV. CONCLUSIONS

In this paper, a new representation for virtual synchronous generators was introduced. While the structure of method is similar to that of conventional VSG methods, the new method makes it possible to accurately capture the dynamics of power

TABLE II
SYSTEM PARAMETERS (SCENARIO 1)

Parameter	Value	Parameter	Value
Grid Voltage		Line inductance values	
V_{RMS}	415V	L_{line1}	0.01mH
Loads values		L_{line2}	0.1mH
Load1	4.3kW, pf=0.995	L_{line3}	0.02mH
Load2	8.6kW, pf=1	L_{line4}	0.1mH
Load3	2.86kW, pf=0.7	L_{line5}	0.03mH

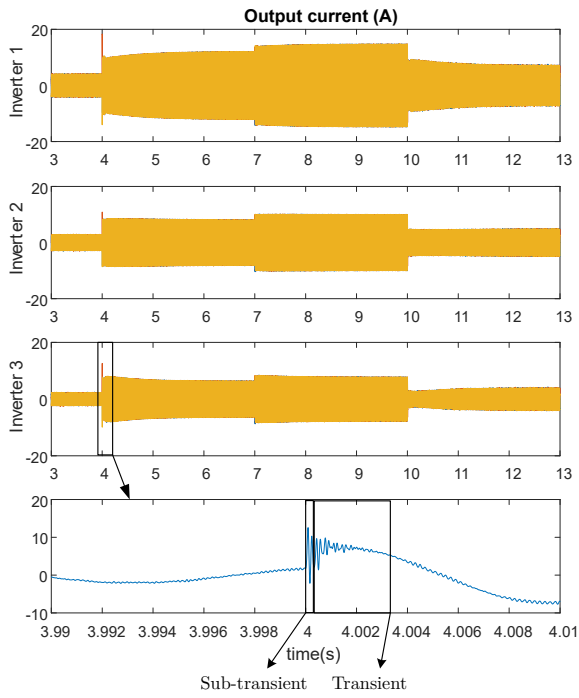


Fig. 5. Output current of the inverters in the first scenario

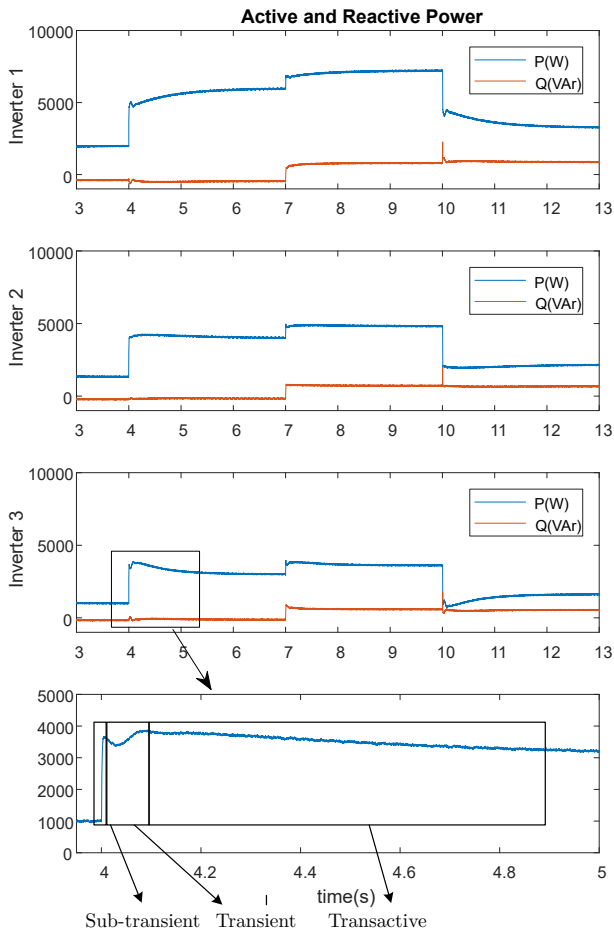


Fig. 6. Active and reactive power of the inverters in the first scenario

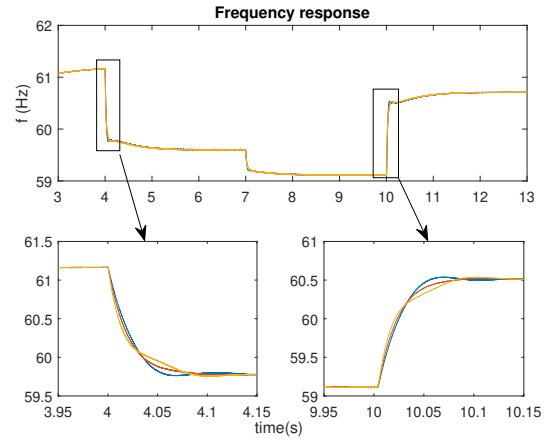


Fig. 7. Frequency response of the inverters in the first scenario

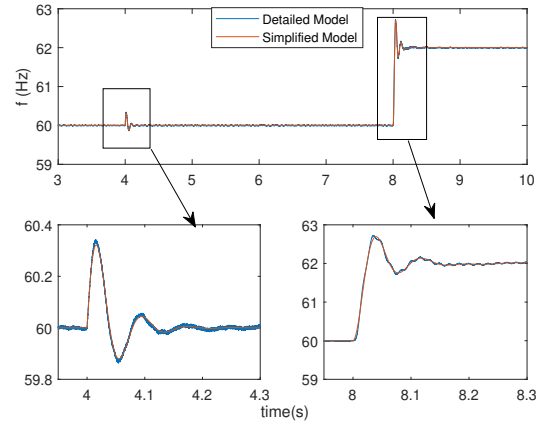


Fig. 8. Dynamic response of the first inverter during transients: detailed model vs. simplified model

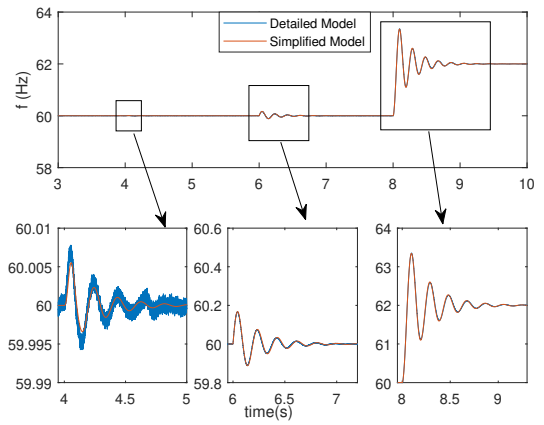


Fig. 9. Dynamic response of the second inverter during transients: detailed model vs. simplified model

TABLE III
SYSTEM PARAMETERS (SCENARIO 2)

Parameter	Value	Parameter	Value
Inductance values		Effective Inductance	
L_{line1}	0.3mH	L_{eff1}	27.8mH
L_{line2}	0	L_{eff2}	29.7mH
L_{line3}	0.5mH	$L_{eff,grid}$	0.81mH
L_{grid}	0.3mH		
Virtual capacitor		Virtual Admittance	
C_{v1}	5mF	L_{va}	10mH
C_{v2}	30mF	R_{va}	20mΩ
R_{v1}	4Ω		

swing with a more simplified circuit. The simulation results showed the feasibility of the method and accuracy of the model for basic studies. It was shown that the method could be extended to higher number of inverters. In large system stuides where there are huge number of grid-forming inverters connected together, this simplified model could be leveraged to study the inverters behavior and their interaction with one another and with the grid.

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